Interpretable MRI-Based Biomarkers for Alzheimer's Disease Classification

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Abstract. Alzheimer's disease (AD) is a progressive neurodegenerative disorder marked by structural brain changes detectable through neuroimaging, particularly Magnetic Resonance Imaging (MRI), which can reveal the extent of atrophy in cortical and subcortical regions. We propose an ensemble of linear models for AD classification, combining regression and classification techniques for a novel methodology to identify potential biomarkers from MRI data, termed Apparent Brain Features (ABF). These biomarkers represent morphological brain regions automatically selected to optimise classification accuracy while preserving interpretability. Unlike deep learning or other nonlinear methods, our approach maintains the anatomical semantics of the input space. A key innovation is a feature score that quantifies the influence of each selected morphological region on classification, enabling both diagnostic utility and neuroscientific insights. We validate our approach on MRI scans from 1990 subjects gathered from four publicly available repositories: ADNI, AIBL, PPMI, and IXI. The results show that our ensemble methodology achieves high classification accuracy while offering an interpretable framework for assessing the role of brain morphology in AD. A systematic selection and evaluation of brain regions can provide a transparent and clinically relevant tool, supporting both computational neuroscience research and practical diagnostic applications.

Keywords: Alzheimer's Disease · Apparent Brain Features · Ensembles · Interpretability

1 Introduction

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by progressive cognitive decline and structural brain atrophy. Early detection and accurate classification of AD, particularly in differentiating it from normal brain aging, remain critical challenges in neuroscience and clinical practice [1]. Machine learning (ML) has become an important tool for aiding diagnosis by leveraging neuroimaging data to detect subtle structural changes. However, many existing ML approaches, while achieving high classification accuracy, rely on complex nonlinear black-box models that lack interpretability [2]. This trade-off between accuracy and interpretability limits their clinical applicability, as understanding the decision-making process is essential for trust and validation in medical settings [2]. Consequently, identifying reliable and interpretable biomarkers that can distinguish pathological deviations from normal brain aging is essential for early diagnosis and timely intervention.

To address this challenge, we propose an ensemble of linear models that integrates regression and classification techniques to derive novel, ML-based biomarkers for Alzheimer's disease. The method systematically selects and leverages morphological brain regions as predictive features. We define these biomarkers as *Apparent Brain Features* (ABFs) — regions whose measurements are highly informative for classification. Our methodology is based on a targeted feature selection procedure aimed at optimising classification performance while maintaining the anatomical semantics of the input space.

ABF models estimate measurements of Regions of Interest (ROIs), such as cortical thickness, surface area, and volume, using regression models built from other ROI measurements. Unlike traditional imputation, this approach repurposes regression to extract features that are most informative for classification. Here, regression accuracy is secondary to the ultimate goal of maximising classification performance, with the focus on deriving brain features that enhance predictive accuracy.

ABF models can be used individually or in ensembles, and offer a significant advantage. They retain intrinsic interpretability, allowing researchers to explain predictions with specific brain regions by maintaining the morphological semantics of the input space. This is an important advantage over deep learning and other black-box methods, which transform data into internal representation spaces that lack direct interpretability. In addition, a feature scoring system is introduced to quantify the contribution of ROIs to the prediction outcome.

The effectiveness of the proposed approach is validated through a comparative experimental analysis, demonstrating its potential as a reliable and interpretable tool for AD classification. The main contributions of this work include:

- a systematic method for estimating measurements of brain regions of interest (Apparent Brain Features),
- the adoption of an inductive bias through feature selection to improve classification accuracy,
- the design of a workflow that exclusively uses linear models to maintain domain semantics.
- the definition of a feature scoring index to measure the contribution of each morphological region to the classification outcome, and
- an experimental validation based on multi-source data.

This paper is structured as follows. Section 2 provides a review of related work and of ensembles of linear models. Section 3 provides a description of the dataset. Section 4 explains the proposed methodology, including the feature selection process, linear models concatenation, model selection and the final ensemble model. Section 5 presents the experimental analysis. Finally, Section 6 provides some conclusions and outlines directions for future research.

2 Related Work

Machine learning regression models have been used to predict brain age as a biomarker for AD, where a higher predicted brain age relative to chronological age may indicate cognitive decline [3, 4]. Common approaches include linear regression, support vector regression, Gaussian processes, deep learning, and ensemble methods such as XGBoost and random forests. These models are typically trained on healthy individuals to learn normative age-related brain changes and identify deviations associated with AD.

In this work, we extend and generalise this approach beyond brain age to other brain features by constructing an ensemble of linear models targeting the most predictive regions with respect to a specific neurodegenerative disease.

2.1 Brain age

Brain age estimation is an important technique for detecting abnormal aging patterns associated with neurodegenerative diseases. Brain Age Gap Estimation (BrainAGE) [3, 4] is a prominent approach that estimates an individual's brain age using Relevance Vector Regression (RVR) and Support Vector Regression (SVR). It applies Voxel-Based Morphometry (VBM) and Principal Component Analysis (PCA) to reduce the large number of voxels to a feature space with lower dimensionality. The difference between the estimated brain age and chronological age, termed the brain age gap, serves as a biomarker for detecting and tracking neurodegenerative progression. BrainAGE has been used to predict the conversion from mild cognitive impairment (MCI) to AD [5] and to classify AD [6]. Other methods, such as Brain Estimated Age Difference (Brain-EAD) [7] and DeepBrainNet [8], aim to minimise age regression residuals and maximize correlation between predicted and actual age in healthy individuals. However, these approaches focus exclusively on global brain age, offering an overall measure of age-related deviation without identifying region-specific changes that may be critical for understanding.

2.2 Ensembles of linear models

Ensembles of linear models [9–11] have been used to improve classification accuracy in high-dimensional datasets. In some cases, these ensemble models have been used specifically to identify biomarkers [12]. For example, the methodology in [10] constructs multiple logistic regression models, each trained on different subsets of features or data partitions. By aggregating the predictions from these diverse models, the ensemble aims to improve robustness and performance, particularly in scenarios with limited sample sizes.

The present study introduces a novel ensemble strategy for classifying AD using region-specific linear models. In this approach, each model focuses on a specific brain region (ROI), applying linear regression and logistic regression to predict diagnosis. This approach improves robustness through efficient ensemble voting and supports the identification of structural biomarkers, aiming for a semantically meaningful and interpretable diagnosis.

3 Data preprocessing

This study uses T1-weighted structural magnetic resonance images of 2001 subjects (931 males and 1070 females), sourced from four publicly available repositories: ADNI (Alzheimer's Disease Neuroimaging Initiative), AIBL (Australian Imaging Biomarker & Lifestyle Study of Aging), IXI (Information eXtraction from Images) and PPMI (Parkinsons Progression Markers Inititative). Images of cognitively normal (CN) and AD subjects from ADNI and IXI are selected. Images from IXI and PPMI are included to increase the number of CN subjects and to improve generalization of the models. In total, the data set includes 466 subjects affected by AD and 1535 CN subjects. For subjects with multiple scans, only screening and baseline scans were used to support early diagnosis. Table 1 provides the distribution of subjects used in this study.

Gender	**Source**	**Group**	**Subjects**	**Age**			
				Mean	SD	Min	Max
Male	ADNI	AD	213	75.82	7.86	56.3	90.4
	ADNI	CN	317	74.20	6.86	56.0	90.3
	AIBL	AD	32	74.65	9.07	58.9	88.4
	AIBL	CN	199	74.36	7.83	54.6	88.8
	IXI	CN	90	65.30	7.27	55.09	84.2
	PPMI	CN	80	62.30	6.07	56.09	83.2
Female	ADNI	AD	177	74.29	8.07	56.2	91.0
	ADNI	CN	398	72.10	6.84	56.3	89.9
	AIBL	AD	44	75.27	7.91	56.5	88.4
	AIBL	CN	272	74.44	7.31	55.5	88.0
	IXI	CN	143	65.10	6.31	55.22	86.3
	PPMI	CN	36	64.30	8.27	54.09	86.2

Table 1: Distribution of the 2001 subjects included in this study.

MRI images were preprocessed using FreeSurfer 6.0 [13] for tasks like skull stripping, registration, segmentation, and cortical thickness estimation, resulting in 446 numerical features representing various brain regions. After data cleaning, the final dataset included 403 features and the subject's age and gender.

In addition, used used iForest [14] with Tukey's method [15] for efficient high-dimensional outlier detection and removal, which identified seven female outliers and four male outliers, reducing the dataset to 1990 subjects. Finally, this study applied intracranial volume (ICV) normalization [16] to account for brain size variability and to reduce confounding effects, improving the sensitivity of statistical comparison. We adopted a linear regression approach for ICV normalisation [16] to adjust ROI measurements, ensuring that regional variations reflect biological or pathological differences.

4 Ensemble of Apparent Brain Features

The apparent brain feature (ABF) method estimates the measurements of individual brain structures, similarly to the apparent brain age that reflects the subject's age. Using a leave-one-out strategy, regression models are built by predicting each brain region from the others. Each ABF model is a combination of linear regression to estimate the feature value and logistic regression to predict the class based on the predicted and actual values of the feature. Finally, an ensemble of the ABF models is built.

4.1 Feature Selection

A filter feature selection method was applied using mutual information (MI) to rank all input features based on their dependency with the target class, selecting the top 10% (40 features). A reduced feature set was then determined by a biased forward feature selection (BFSS) process [17] with a wrapper approach that includes linear regression and logistic regression models. This feature selection phase is applied to each ROI measurement (potential biomarker) and results in a subset of predictors for each target feature j: $F_j = \{f_i\}$ $(1 \leq i \leq N_j \leq 40)$.

The next section describes the single-feature models built on this target-specific feature selection approach.

4.2 ABF models

The ABF linear regression model for each ROI feature j is:

$$\hat{f}_j = a_0 + \sum_{i=1}^{N_j} a_i \cdot f_i, \tag{1}$$

where $\hat{f}_j$ is the regression prediction, N_j is the number of selected predictors f_i for f_j and $\{a_i\}$ $(0 \leq i \leq N_j)$ is the set of the $N_j + 1$ linear coefficients.

The logistic regression model is a linear decision boundary between the two classes (AD and CN) in a two-dimensional space, based on the predicted value of the ROI and its actual value. It is used to perform the prediction task by combining linear regression and logistic regression, resulting in a single linear model that can directly perform the classification task from the selected predictors. The linear boundary in the logistic regression is given by:

$$c_0 + c_1 \cdot f_j + c_2 \cdot \hat{f}_j > 0, \forall j. \tag{2}$$

Considering equation (1), the single ABF classifier (2) can be expressed directly in terms of the input features according to:

$$c_0 + c_1 \cdot f_j + c_2 \cdot \left(a_0 + \sum_{i=1}^{k} a_i \cdot f_i \right) > 0, \forall j. \tag{3}$$

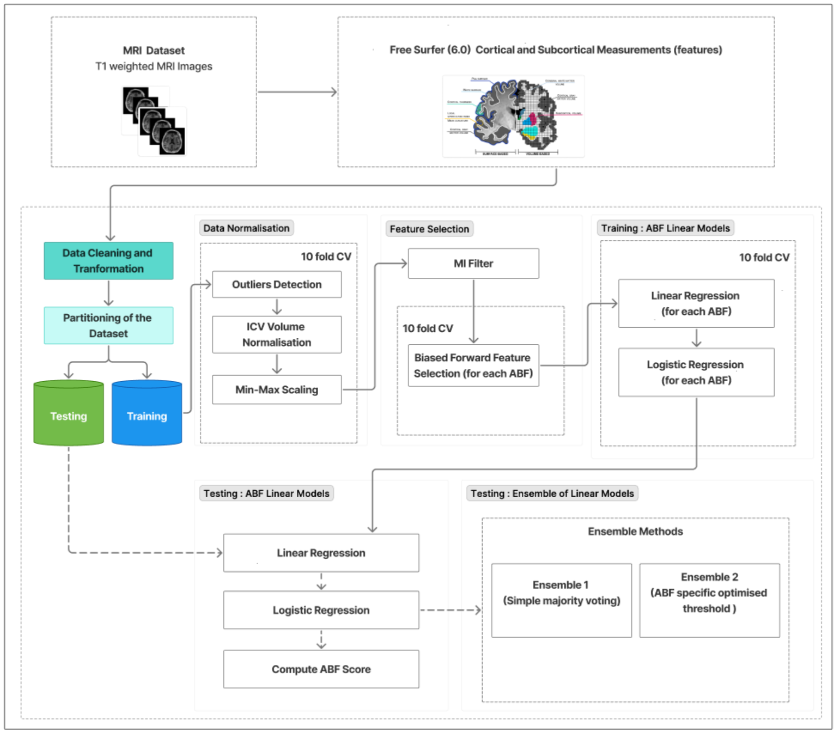


Fig. 1: Overall data workflow of the proposed method.

The above equation 3 can be expressed as

$$\sum_{i=1}^{k} s_{i,j} > 1, \forall j, \tag{4}$$

where $s_{i,j}$ is the ABF score of the predictor f_i for the target feature f_j, and is defined as

$$s_{i,j} = \frac{c_2 a_i f_i}{-(c_0 + c_1 \cdot f_j + c_2 \cdot a_0)}. \tag{5}$$

The ABF score of a predictor feature indicates its contribution to the classification inequality in equation 4, with a low score signifying minimal or no atrophy and a high score suggesting abnormal atrophy. The sum of the feature scores provides the model prediction, while the individual scores provide the interpretability of the model explaining the prediction outcome directly and quantitatively in terms of the ROIs.

The challenge is to identify and select the most relevant target features f_j and the most important predictor features f_i.

Finally, given a set $\{f_j\}$ of target features, an ensemble of their ABF linear classifiers is built to improve robustness and generalisation. Two ensemble approaches have been investigated as briefly described in the following section.

4.3 Linear models optimisation and ensembles

The Apparent Brain Feature (ABF) classifier is a single feature classifier that uses linear and logistic regression to classify samples as CN or AD. The classification rule of a single classifier is based on a threshold that is optimised by cross-validation and a grid search in the range $[0.00, 1.00]$ with a step size of 0.01 to maximise classification accuracy. Two ensemble strategies are devised to combine predictions from the set of ABF classifiers.

- **Ensemble 1:** Each ABF model contributes a binary vote (CN/AD) using their own optimised threshold. A fixed majority voting rule was applied: a test sample is classified as AD if at least 50% of the models predict AD.
- **Ensemble 2:** Similarly to Ensemble 1, each ABF model uses their own optimised threshold for binary voting (CN/AD). However, instead of a fixed majority rule, the ensemble majority voting threshold is also optimised.

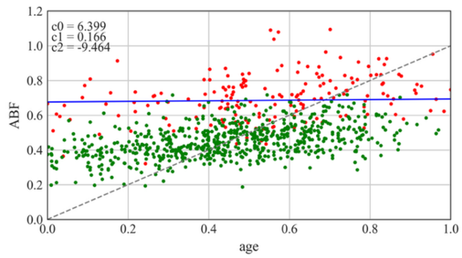
This framework allows each linear model to operate independently, yet contribute collectively, balancing strong individual performance with ensemble robustness. A detailed comparison of the performance of individual models and the two ensemble strategies is presented in the next section.

5 Experimental Analysis

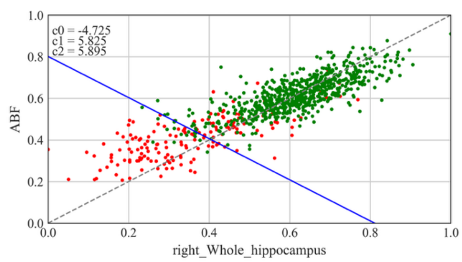
ABF models are trained and tested on structural MRI data extracted from the brain images of AD and CN subjects. Each ABF model consists of a linear regression model and a logistic regression model. Figure 2 illustrates three representative ABF models, each capturing a distinct neuroanatomical phenomenon associated with Alzheimer's disease. The first example in Figure 2a demonstrates that the Apparent Brain Age (ABA) [17] model can serve as a reliable and interpretable biomarker, consistently overastimating the chronological age in AD subjects. This overestimation highlights the model's sensitivity to the neurodegenerative patterns in AD subjects. With a training accuracy of 89% and a testing accuracy of 87%, the ABA model offers a compelling tool for the early diagnosis of AD.

The ABF model for the left whole hippocampus (Figure 2b) shows a good classification performance. For AD subjects, the model shows a reduction in the hippocampus volume, indicating disease-driven atrophy. The whole hippocampus is deemed a highly predictive region for AD classification, achieving 89% training and 90% testing accuracy.

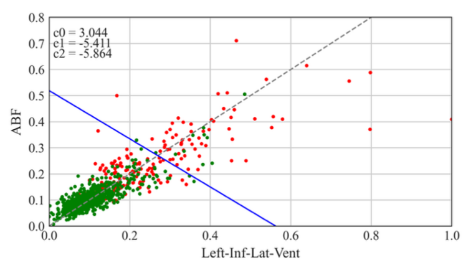
Figure 2c shows the case of the inferior lateral ventricle, whose volume is enlarged in AD subjects. This model achieved 85% accuracy in both training and testing, demonstrating reliability as an indicator of neurodegeneration.



(a) ABF model for ABA, illustrating accelerated brain aging in AD, where predicted age (ABA) exceeds chronological age.



(b) Left whole hippocampus: volume reduction in gray matter region.



(c) Left inferior lateral ventricle enlargement.

Fig. 2: Visualization of ABF models for the training data (left) and test data (right). Red dots represents AD subjects and green dots CN subjects. The blue line is the classification boundary.

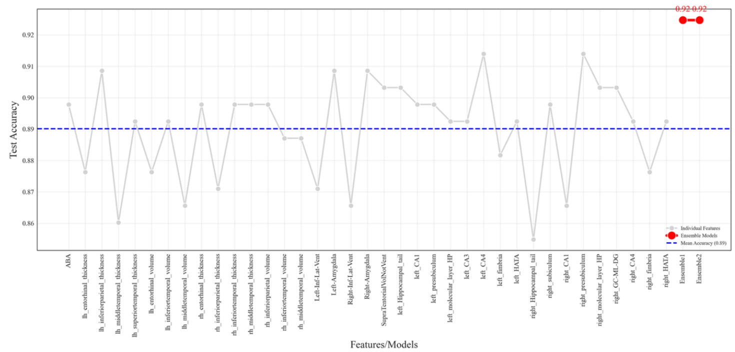
A comparison of the classification performance of the ensembles and the single-feature ABF models for a single hold-out test is shown in Figure 3. Ensemble methods show a strong accuracy, proving their reliability as predictive tools. Single-feature ABFs, such as the left_CA4 and right_presubiculum in male subjects and lh_entorhinal volume, right_hippocampal_tail, left_molecular_layer _HP, and right_subiculum in female subjects, achieved high accuracy comparable to that of the ensemble models. Ensemble models perform better in male subjects, while specific ABFs excel in females, indicating gender-related variability in predictive brain features for Alzheimer's disease. Despite the strong performance of some individual ABFs, the ensemble models demonstrated greater stability and generalization.

Finally, to benchmark performance, we compared the method against Support Vector Machines (SVM) and Deep Neural Networks (DNN), as shown in Table 2. The estimation of model accuracy involved 10-fold cross-validation (CV) on the training set and a repeated (10 times) hold-out method on a test set, as shown in Figure 1. For SVM, an RBF kernel with default parameters was used. SVM without feature selection achieved a training CV accuracy of 96% (male) and 98% (female), and hold-out (HO) test accuracy of 89% (male) and 90% (female), which may indicate overfitting. The DNN model used two hidden layers, batch normalisation, dropout (0.3), and Adam optimiser. DNN without feature selection achieved training CV accuracy of 93% for both genders and test accuracy of 88% (male) and 87% (female). The inclusion of the MI filter for feature selection slightly improved the test accuracy of the DNN and SVM models. These models lack interpretability and would require additional explainability methods to investigate which regions of the brain influence their predictions. Both Ensemble methods achieve comparable or slightly better results on the test set, demonstrating the effectiveness of the proposed ABF methodology.

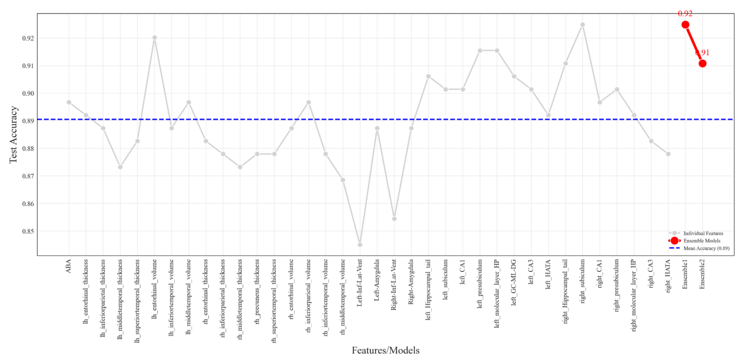
Additionally, in order to illustrate model interpretability Figure 4 shows the ABF scores of a single feature model for two subjects, one for each class. The ABF scores allow to investigate the relative contributions of the predictors to the classification outcome provided by the model. In the CN case, the contributions of all the predictors add up to a value (-3.52) that determines a correct CN prediction. In the AD case, the overall sum of the scores (1.81) is above the threshold (equation 4) correctly predicting the class AD. It is possible to identify which predictors most contribute to the classification output, which in this case are: *left entorhinal thickness*, *right presubiculum*, *left granule cell and molecular layer of the Dentate Gyrus (GC ML DG)*, and *right CA3 of the Hippocampus*.

6 Conclusions

We presented an interpretable MRI-based method for AD classification, using an ensemble of linear models that combine regression and classification models. The proposed Apparent Brain Features (ABFs) are anatomically meaningful biomarkers that deliver high accuracy while preserving interpretability. Evaluated on 1990 subjects, the Ensemble models achieved strong performance and identified clinically relevant cortical areas. Despite these strengths, our approach has certain limitations. It relies exclusively on structural MRI data, omitting cognitive and other risk factors such as APOE4, and does not account for intermediate disease stages such as mild cognitive impairment (MCI). Future work will incorporate longitudinal data to improve early detection and monitoring over time and include MCI subjects to investigate AD transitions. In addition, the applicability of the method can be extended to other neurodegenerative diseases.



(a) Male subjects



(b) Female subjects

Fig. 3: Single hold-out test accuracy of the ABA model [17], the single-feature models (ABF), and the two proposed ensemble methods, stratified by gender.

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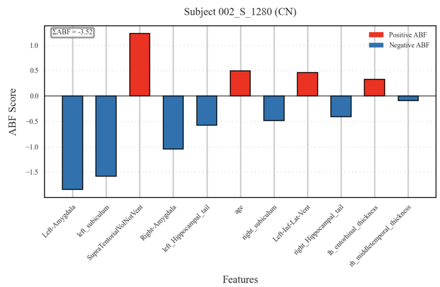
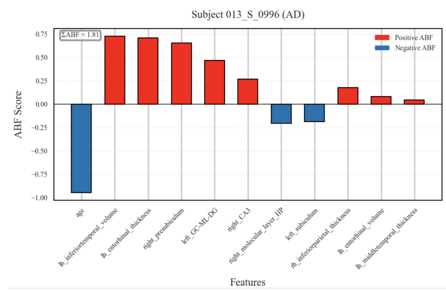
References

1. S. Safiri, A. Ghaffari Jolfayi, and Fazlollahi, “Alzheimer’s disease: a comprehensive review of epidemiology, risk factors, symptoms diagnosis, management, caregiving, advanced treatments and associated challenges,” *Frontiers in Medicine*, vol. Volume 11 - 2024, 2024.
2. S. A. Martin, F. J. Townend, F. Barkhof, and J. H. Cole, “Interpretable machine learning for dementia: a systematic review,” *Alzheimer’s & Dementia*, vol. 19, no. 5, pp. 2135–2149, 2023.



(a) ABF - CN



(b) ABF - AD

Fig. 4: ABF scores of linear models for the Hippocampal subfield CA3 of a CN subject and for the inferior lateral ventricle of an AD subject.

Gender	**Model**	**Feature Selection**	**Training Accuracy (CV)**	**Test Accuracy (HO)**
Male	DNN	-	0.93	0.88
	DNN	MI Filter	0.93	0.90
	SVM	-	0.96	0.89
	SVM	MI Filter	0.96	**0.91**
	Ensemble1	MI Filter, BFFS	0.90	**0.91**
	Ensemble2	MI Filter, BFFS	0.91	**0.91**
Female	DNN	-	0.93	0.87
	DNN	MI Filter	0.94	0.90
	SVM	-	0.98	0.90
	SVM	MI Filter	0.98	0.91
	Ensemble1	MI Filter, BFFS	0.93	**0.92**
	Ensemble2	MI Filter, BFFS	0.94	**0.92**

Table 2: Model performance comparison stratified by gender: 10-fold cross-validation (CV) was used to estimate the training accuracy, 10-time repeated hold-out (HO) for the test accuracy with relative standard deviation below 2.7%.

3. K. Franke, G. Ziegler, S. Klöppel, and C. Gaser, "Estimating the age of healthy subjects from t1-weighted MRI scans using kernel methods: Exploring the influence of various parameters," *NeuroImage*, vol. 50, no. 3, pp. 883–892, Apr. 2010.
4. K. Franke and C. Gaser, "Ten years of brainage as a neuroimaging biomarker of brain aging: What insights have we gained?" *Frontiers in Neurology*, vol. 10, p. 789, 2019.
5. C. Gaser, K. Franke, S. Klöppel, N. Koutsouleris, and H. Sauer, "BrainAGE in mild cognitive impaired patients: Predicting the conversion to alzheimer's disease," *PLoS ONE*, vol. 8, no. 6, Jun. 2013.
6. K. Franke and C. Gaser, "Dementia classification based on brain age estimation," *Proc MICCAI Workshop Challenge on Computer-Aided Diagnosis of Dementia Based on Structural MRI Data*, pp. 48–54, 2014.

7. I. Beheshti, S. Mishra, D. Sone, P. Khanna, and H. Matsuda, "T1-weighted MRI-driven brain age estimation in alzheimer's disease and parkinson's disease," *Aging and disease*, vol. 11, no. 3, p. 618, 2020.
8. V. M. Bashyam, G. Erus, J. Doshi, M. Habes, I. M. Nasrallah, M. Truelove-Hill, D. Srinivasan, L. Mamourian, R. Pomponio, Y. Fan, L. J. Launer, C. L. Masters, P. Maruff, C. Zhuo, H. Völzke, S. C. Johnson, J. Fripp, N. Koutsouleris, T. D. Satterthwaite, D. Wolf, R. E. Gur, R. C. Gur, J. Morris, M. S. Albert, H. J. Grabe, S. Resnick, R. N. Bryan, D. A. Wolk, H. Shou, and C. Davatzikos, "MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14 468 individuals worldwide," *Brain*, vol. 143, no. 7, pp. 2312–2324, Jun. 2020.
9. S. Reid and G. Grudic, "Regularized linear models in stacked generalization," in *Multiple Classifier Systems*, J. A. Benediktsson, J. Kittler, and F. Roli, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 112–121.
10. H. M. Noha Lim, Hongshik Ahn and J. J. Chen, "Classification of high-dimensional data with ensemble of logistic regression models," *Journal of Biopharmaceutical Statistics*, vol. 20, no. 1, pp. 160–171, 2009.
11. M. Ijaz, Z. B. Asghar, and A. Gul, "Ensemble of penalized logistic models for classification of high-dimensional data," *Communications in Statistics - Simulation and Computation*, vol. 50, pp. 2072 – 2088, 2019.
12. K. Gardiner, X. Zhang, and L. Xing, "Bless: bagged logistic regression for biomarker identification," *Frontiers in Genetics*, vol. 15, p. 1336891, 2024.
13. B. Fischl, D. H. Salat, E. Busa, M. Albert, M. Dieterich, C. Haselgrove, A. van der Kouwe, R. Killiany, D. Kennedy, S. Klaveness, A. Montillo, N. Makris, B. Rosen, and A. M. Dale, "Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain," *Neuron*, vol. 33, no. 3, p. 341–355, 2002.
14. F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining*, 2008, pp. 413–422.
15. C. M. Salgado, C. Azevedo, H. Proença, and S. M. Vieira, "Noise versus outliers," in *Secondary Analysis of Electronic Health Records*. Springer International Publishing, 2016, pp. 163–183.
16. R. L. Buckner, D. Head, J. Parker, and A. F. Fotenos, "A unified approach for morphometric and functional data analysis in young, old, and demented adults using automated atlas-based head size normalization: Reliability and validation against manual measurement of total intracranial volume," *NeuroImage*, vol. 23, no. 2, pp. 724–738, 2004.
17. A. Varzandian, M. A. S. Razo, M. R. Sanders, A. Atmakuru, and G. Di Fatta, "Classification-biased apparent brain age for the prediction of alzheimer's disease," *Frontiers in Neuroscience*, vol. 15, 2021.